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House Sales Management System

EDS6397 – Database Management Tools

Group – 1

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1. Overview:

In an era where digital transformation is reshaping thriving industries like the real estate market, there is a constant increasing demand for efficiency, security and user-friendly systems. This project addresses the need and demand by building a the Real Estate Property Management System (REPMS), primarily to facilitate the sale and purchase of residential properties, manage property listings, and provide users with tools for property search and selection – all through an integrated interface. Given the growing rates of property markets and the growing complexity of transactions, the need for a digital solution to manage, facilitate and maintain a secure audit trail of all user activities arises. Our system supports multiple user roles: buyers, sellers, real estate agents, and administrators with Buyer seeking to explore and look at potential properties to purchase; Seller putting out their property information and details to be sold; Agents who could facilitate the property buying and selling as a third part bridge; Administrators who maintain and manage the platform.

Mainly for REPMS, we aimed to create a structured relational database accounting for efficient organization, retrieval, and management of critical information, like property details, user profiles, transactional data, and communication logs. We wanted to bridge the gap between property sellers, buyers, and renters by providing a centralized interface that supports the entire real estate transaction lifecycle.

Key modules of this project include property management, appointment scheduling, offer and rental interest tracking, secure payment recording, favorites management, chat communication between users, support ticketing for queries and complaints, and audit logging for enhanced transparency.

We also implemented this work with scalability in mind ensuring that system is adaptable to the evolving needs of property businesses, from small agencies to large-scale real estate firms. Stakeholders get to handle high volumes of listings, transactions, and communications with ease and efficiency. Ultimately, the Real Estate Property Management System provides a modern, reliable, and data-driven foundation for transforming how properties are managed, marketed, and transacted in today's digital world.

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2.Business Rules:

2.1. User

- Each user must have a unique email address for identification and login purposes.
- Passwords must be securely stored using a strong hashing algorithm.
- The role field must clearly define user access (Buyer, Seller, Agent, Admin), and user roles control permissions across the system.

2.2. Property Listings Table (Property Management)

- Only users with Seller or Agent role can Create , Modify and Delete property listings.
- Each property must have a valid user_id from the user table, ensuring ownership is traceable.
- Each Property must have a valid property_status (Available, Hold and Sold).

2.3. Property Image Table (Property Images)

- Each property should have at least one associated image (image_url).
- The propertyimage_id must link to an existing entry in property_listings.

2.4. Appointment Table (Appointments Scheduling)

- Only registered users can schedule appointments (linked via user_id).
- Appointments can be scheduled only for properties with status = 'available'.
- Seller/Agent can Accept or Reject the appointments.

2.5. Offer Table (Offer Management)

- Only users with Buyer and Agent roles can submit offers on properties.
- A Buyer/Agent can submit only one active offer per property.
- When an offer is accepted, all other offers on that property must be automatically rejected.

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2.6. Payment Table (Payments and Transaction Flow)

- Payments are allowed only for offers with status = ‘Hold'.
- Upon successful payment, the property status must update to 'Sold'.
- The amount_paid should be validated to match the Seller accepted offer_price.

2.7. Favorite Table (Favorites Functionality)

- Only registered users can mark properties as favorites.
- A property can be favorited only once by the same user.

2.8. Tickets Table (Support System / Help Center)

- Each ticket must be linked to a valid user_id.
- Ticket status must be either 'Open' or 'Resolved’.
- Once a ticket is resolved, a resolution message must be recorded in the resolution field.

2.9. Audit Log Table (Action Tracking and Monitoring)

- Every action performed by a user must be recorded with action_type, action_date, and a description.
- Linked via user_id to identify which user performed the action.
- The description field must clearly explain the action taken (e.g., 'Offer submitted', 'Payment processed’).

2.10. Rental Contracts Table (Rental Management)

- Each rental contract must reference a valid property_id.
- Only one active rental contract is allowed per property at any given time.
- Security deposit and available_date must be valid.

2.11. Rental Interest Table (Rental Interest and Payment Status)

- Each rental interest must be associated with a valid buyer and property.
- The status should be managed between 'Pending', 'Accepted', or 'Rejected'.
- If payment is completed (payment_status = 'Paid’) and (property_status=‘Sold’) and the property will get displayed under his properties.

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3.Entity-Relationship Diagram (ERD):

Our Entity-Relationship Diagram (ERD) includes **11 entities**, each with key attributes and relationships. It shows how users, properties, offers, payments, and other components are connected, ensuring organized and efficient data management.

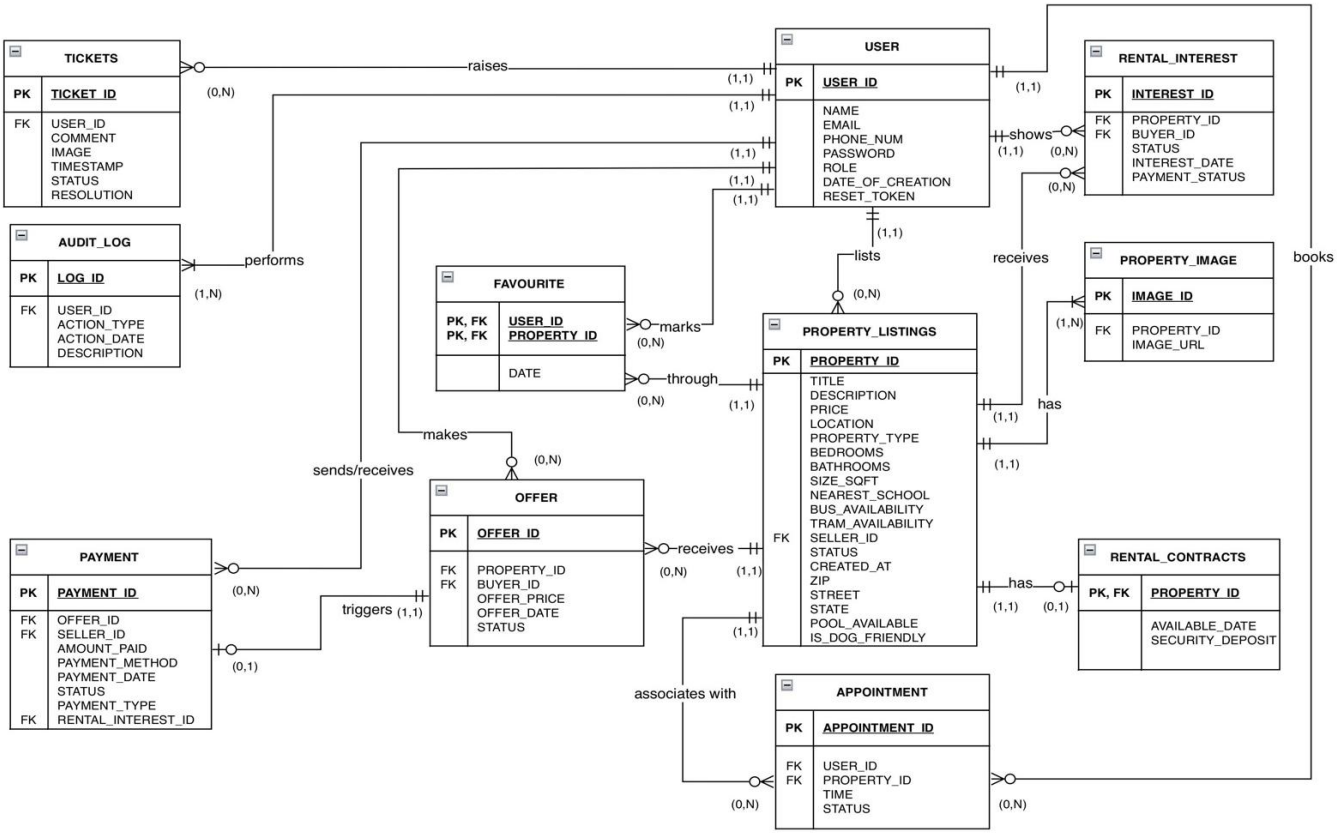

Figure: Entity-Relationship Diagram (ERD) for the Real Estate Property Management System

Each entity in the ERD is equipped with a **Primary Key (PK)** to uniquely identify its records, and necessary **Foreign Keys (FK)** are used to establish strong relational integrity between tables. The major entities include USER, PROPERTY_LISTINGS, OFFER, RENTAL_INTEREST, PAYMENT, FAVOURITE, PROPERTY_IMAGE, TICKETS, AUDIT_LOG, RENTAL_CONTRACTS, and APPOINTMENT.

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4. Relational Diagram:

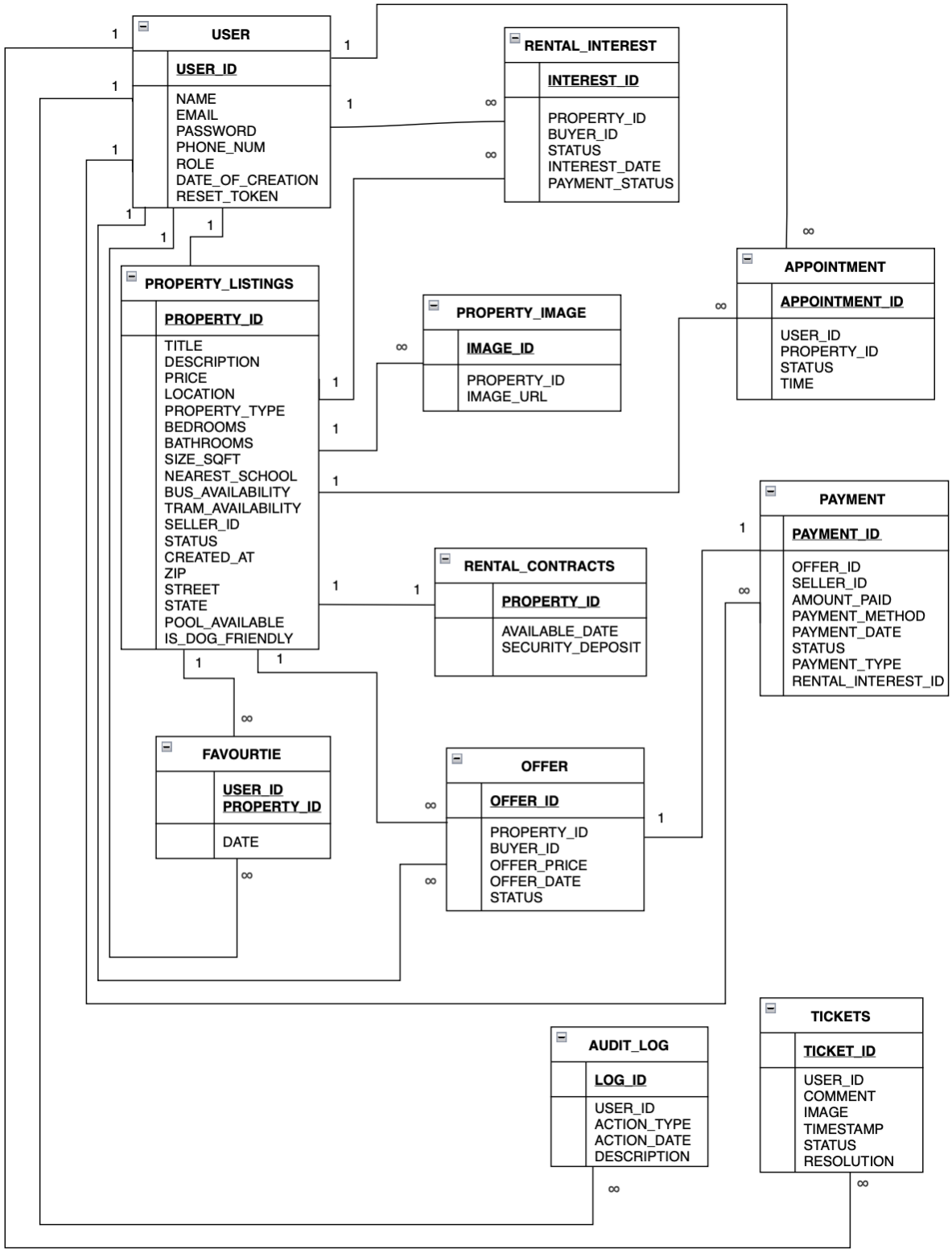

Figure: Relational Diagram for the Real Estate Property Management System

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5. Data Dictionary:

Entities	Attributes	Datatype	IS NULL (Y/N)	PK / FK
USER	**USER_ID**	INT(11)	N	PK
	NAME	VARCHAR(255)	N	
	EMAIL	VARCHAR(255)	N	
	PHONE_NUM	VARCHAR(255)	N	
	PASSWORD	VARCHAR(255)	N	
	ROLE	VARCHAR(255)	N	
	DATE_OF_CREATION	DATETIME	N	
	RESET_TOKEN	VARCHAR(255)	Y	
PROPERTY_LISTINGS	**PROPERTY_ID**	INT(11)	N	PK
	TITLE	VARCHAR(255)	N	
	DESCRIPTION	TEXT	Y	
	PRICE	DECIMAL(15,2)	N	
	LOCATION	VARCHAR(255)	Y	
	PROPERTY_TYPE	VARCHAR(100)	Y	
	BEDROOMS	INT(11)	Y	
	BATHROOMS	INT(11)	Y	
	SIZE_SQFT	INT(11)	Y	
	NEAREST_SCHOOL	VARCHAR(255)	Y	
	BUS_AVAILABILITY	ENUM('YES', 'NO')	N	
	TRAM_AVAILABILITY	ENUM('YES', 'NO')	N	
	SELLER_ID	INT(11)	Y	FK
	STATUS	ENUM('AVAILABLE', 'SOLD', 'HOLD')	Y	
	CREATED_AT	TIMESTAMP	N	
	ZIP	VARCHAR(100)	Y	
	STREET	VARCHAR(255)	Y	
	STATE	VARCHAR(100)	Y	
	POOL_AVAILABLE	ENUM('YES', 'NO')	Y	
	IS_DOG_FRIENDLY	ENUM('YES', 'NO')		
APPOINTMENT	**APPOINTMENT_ID**	INT(11)	N	PK
	USER_ID	INT(11)	Y	FK
	PROPERTY_ID	INT(11)	Y	FK
	TIME	DATETIME	Y	
	STATUS	VARCHAR(50)	Y	
AUDIT_LOG	**LOG_ID**	INT(11)	N	PK
	USER_ID	INT(11)	N	FK
	ACTION_TYPE	VARCHAR(50)	N	
	ACTION_DATE	DATETIME	N	
	DESCRIPTION	TEXT	N	

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FAVORITE	**USER_ID** **PROPERTY_ID** DATE	INT(11) INT(11) DATE	N N Y	PK,FK PK,FK
OFFER	**OFFER_ID** PROPERTY_ID BUYER_ID OFFER_PRICE OFFER_DATE STATUS	INT(11) INT(11) INT(11) DECIMAL(12,2) DATE ENUM('PENDING', 'ACCEPTED', 'REJECTED', 'SOLD')	N Y Y Y Y Y	PK FK FK
PAYMENT	**PAYMENT_ID** OFFER_ID SELLER_ID AMOUNT_PAID PAYMENT_METHOD PAYMENT_DATE STATUS PAYMENT_TYPE RENTAL_INTEREST_ID	INT(11) INT(11) INT(11) DECIMAL(12,2) ENUM('CREDIT CARD', 'BANK TRANSFER', 'PAYPAL', 'STRIPE') DATE ENUM('COMPLETED', 'PENDING') ENUM('SALE', 'RENTAL') INT(11)	N Y Y Y Y Y Y Y Y	PK FK FK FK
PROPERTY_IMAGE	**IMAGE_ID** PROPERTY_ID IMAGE_URL	INT(11) INT(11) VARCHAR(255)	N N N	PK FK
RENTAL_CONTRACTS	**PROPERTY_ID** AVAILABLE_DATE SECURITY_DEPOSIT	INT(11) DATE DECIMAL(10,2)	N Y Y	PK,FK
RENTAL_INTEREST	**INTEREST_ID** PROPERTY_ID BUYER_ID STATUS INTEREST_DATE PAYMENT_STATUS	INT(11) INT(11) INT(11) ENUM('PENDING', 'ACCEPTED', 'REJECTED') DATETIME ENUM('PENDING', 'PAID')	N N N Y Y Y	PK FK FK
TICKETS	**TICKET_ID** USER_ID COMMENT IMAGE TIMESTAMP STATUS RESOLUTION	INT(11) INT(11) TEXT VARCHAR(255) DATETIME ENUM('OPEN', 'RESOLVED') TEXT	N N Y Y Y Y Y	PK FK

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6.Sample Queries:

During the in-class presentation, the instructor asked us to write and execute some SQL queries.

Below are the queries that we demonstrated:

6.1.Retrieve all houses purchased by a specific buyer.

```
SELECT pl.PROPERTY_ID FROM property_listings pl JOIN offer o ON pl.PROPERTY_ID = o.property_id JOIN payment p ON p.offer_id = o.offer_id
WHERE o.buyer_id = 72 AND p.payment_type = 'Sale' AND p.status = 'Completed';
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

PROPERTY_ID
57
55
56

6.2.Retrieve all houses with more than 3 bedrooms.

```
SELECT PROPERTY_ID,title FROM property_listings WHERE BEDROOMS > 3;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

←T→			PROPERTY_ID	title
Edit	Copy	Delete	45	Elegant 5BHK Penthouse
Edit	Copy	Delete	46	High-End 4BHK Bungalow
Edit	Copy	Delete	47	5BHK Ultra Luxury Villa with Private Pool
Edit	Copy	Delete	49	Sea-Facing 3BHK Penthouse
Edit	Copy	Delete	51	5BHK Ultra Luxury Villa with Private Pools
Edit	Copy	Delete	53	Modern 3BHK Gated Community Villa
Edit	Copy	Delete	55	High-End 4BHK Bungalow
Edit	Copy	Delete	57	Exellent Mansion

6.3.Retrieve the total number of houses in a specific neighborhood.

```
SELECT COUNT(*) AS total_houses FROM property_listings WHERE LOCATION = 'Houston';
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Extra options

total_houses
7

6.4.Retrieve the top real estate agent with the highest number of sales.

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Your SQL query has been executed successfully.

```
SELECT u.user_id, u.name, COUNT(*) AS total_sales FROM user u JOIN property_listings pl ON u.user_id = pl.seller_id JOIN offer o ON
o.property_id = pl.property_id JOIN payment p ON p.offer_id = o.offer_id WHERE u.role = 'Agent' AND p.payment_type = 'Sale' AND p.status
= 'Completed' GROUP BY u.user_id, u.name ORDER BY total_sales DESC LIMIT 1;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Extra options

user_id	name	total_sales
71	kiran	2

7.Sample Pictures:

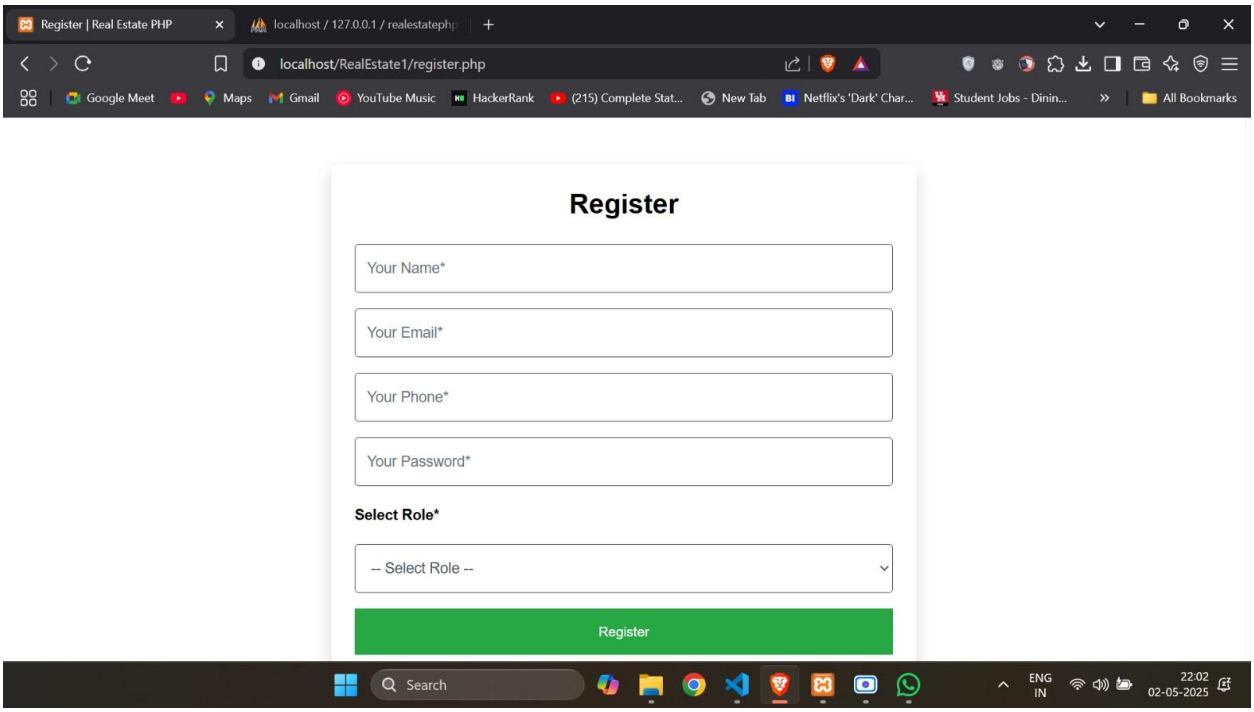

Fig1 : Account Registration

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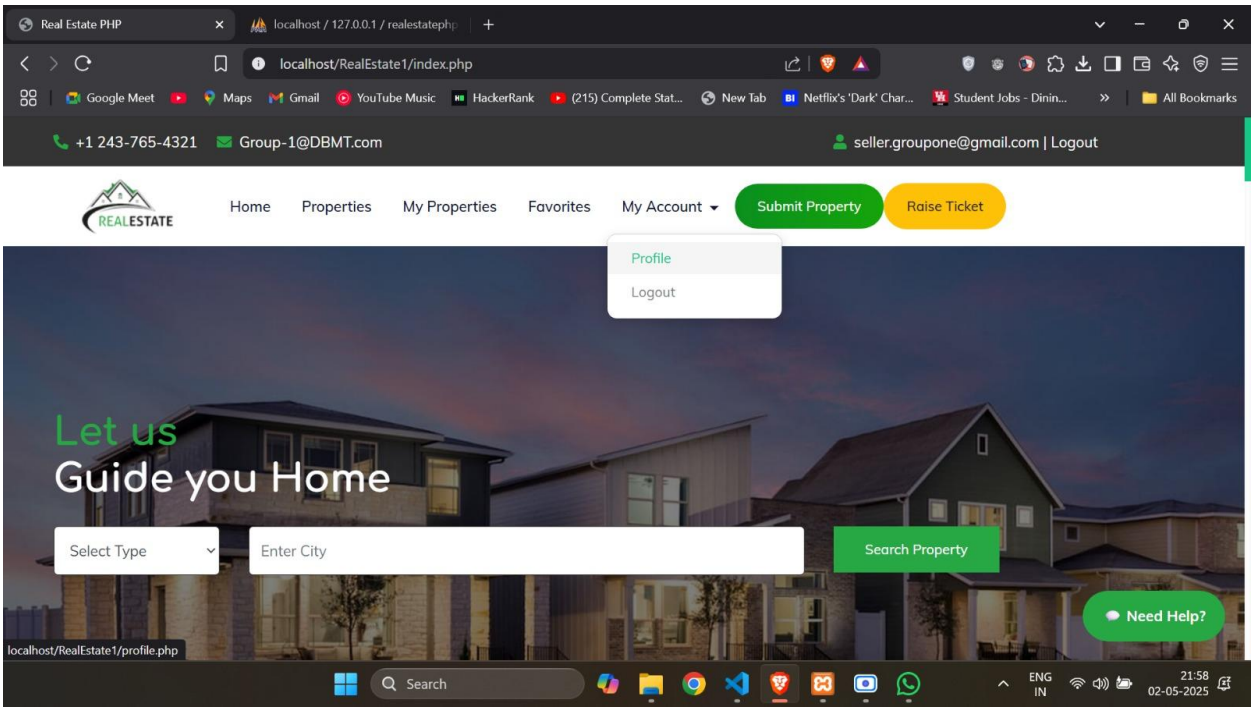

Fig2: Home-Page

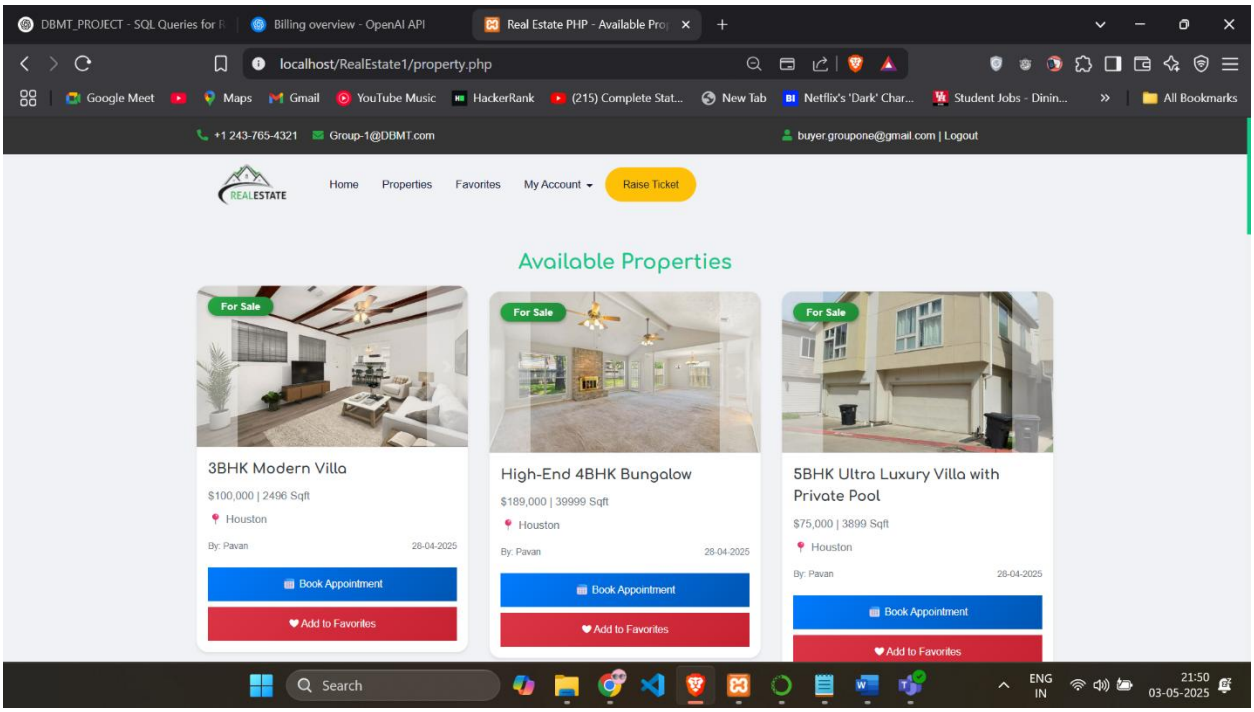

Fig 3: Available Properties

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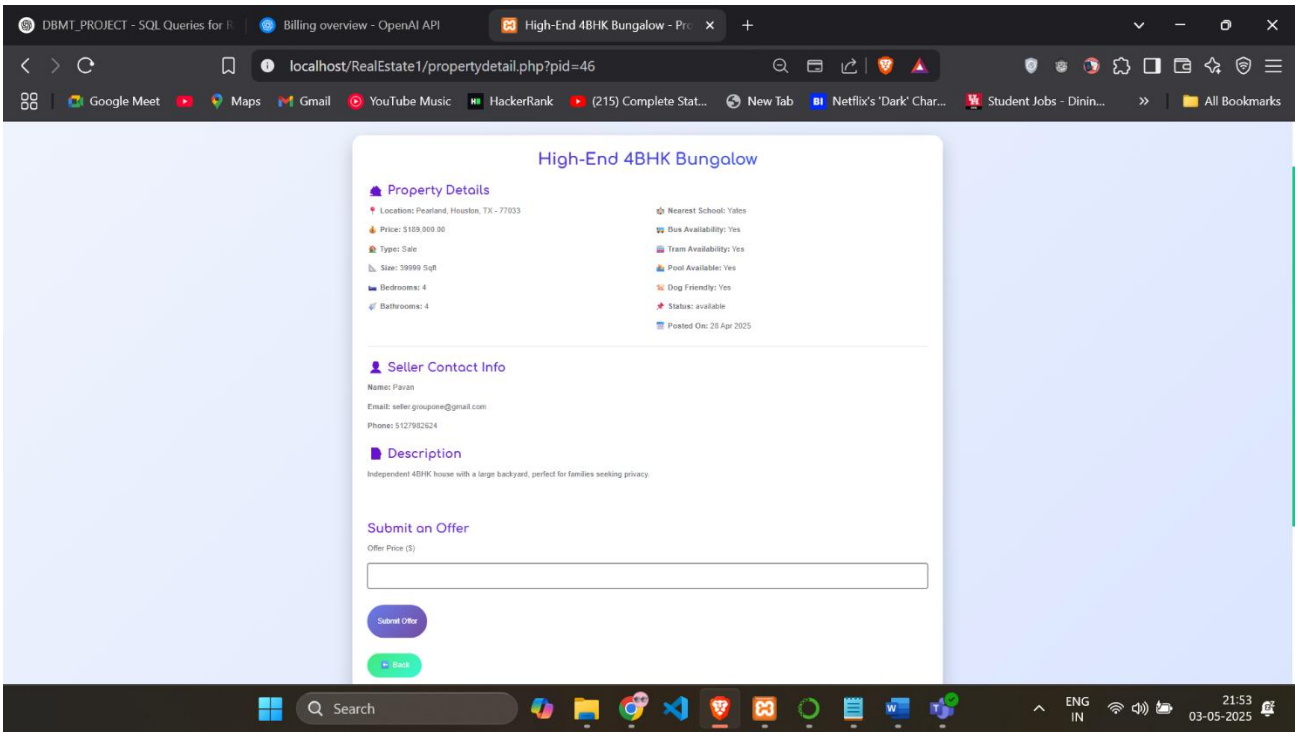

Fig 4: Properties Details

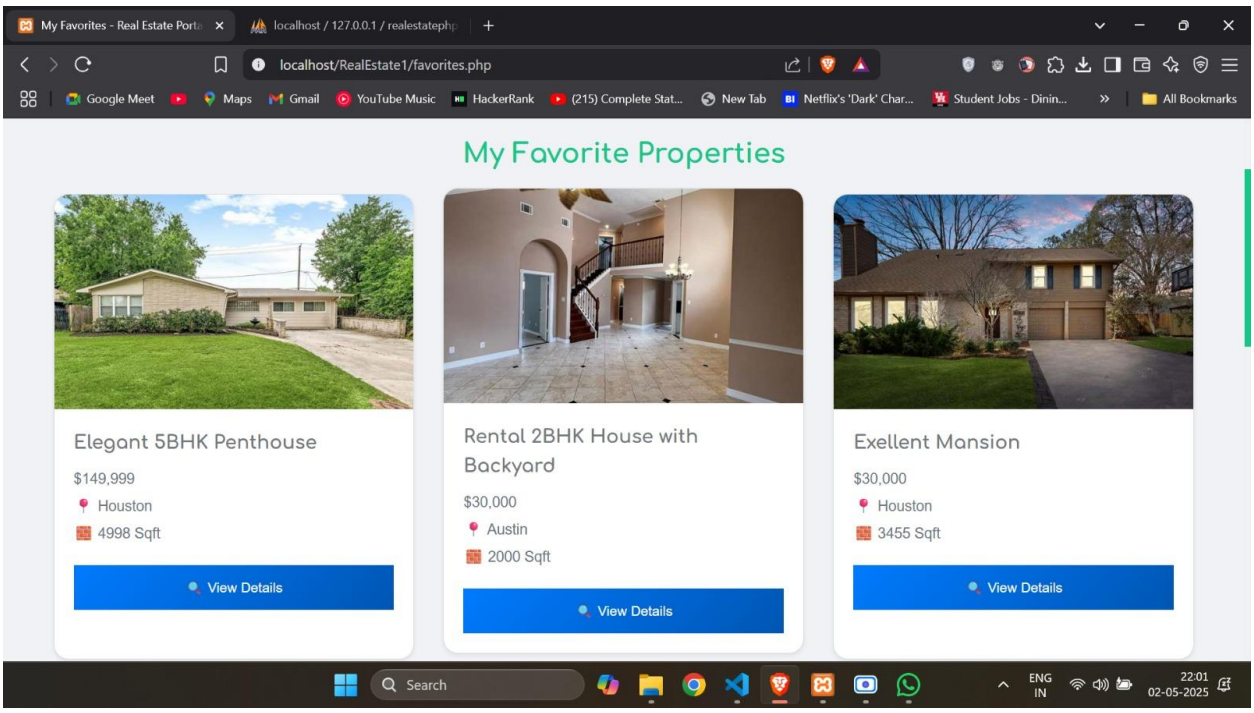

Fig5 : Favorite Properties

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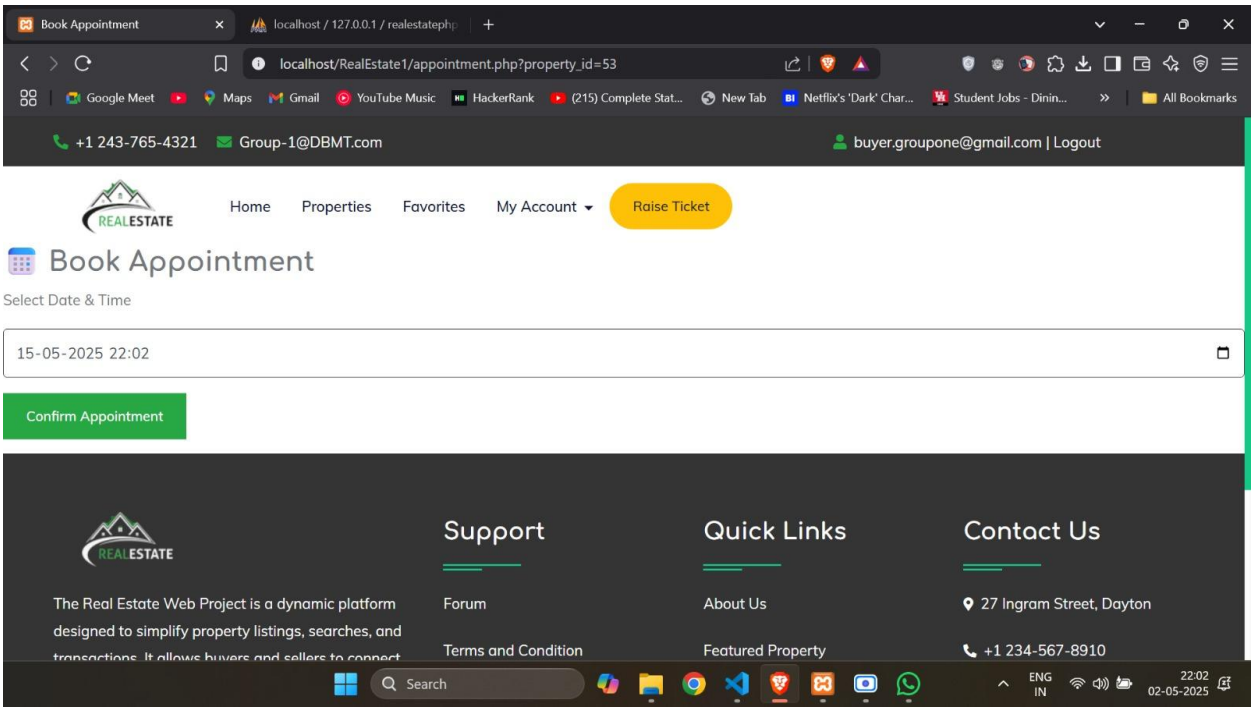

Fig6 : Appointment Booking

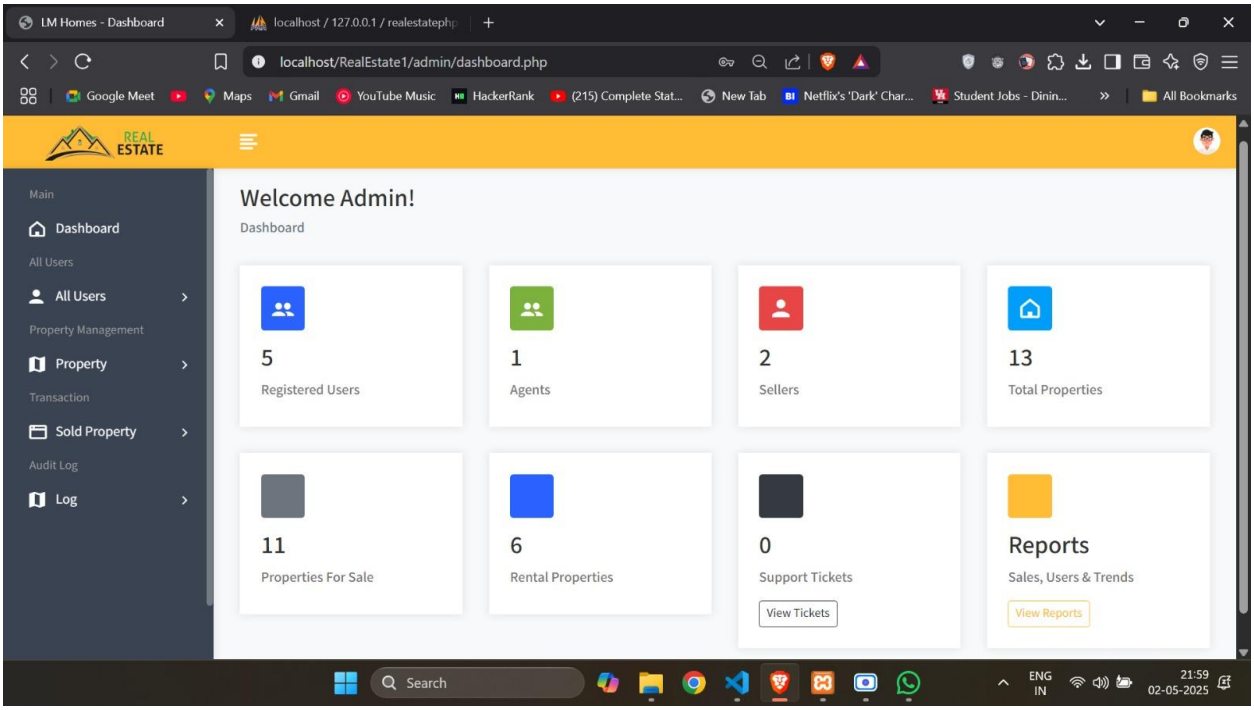

Fig 7 : Admin Page

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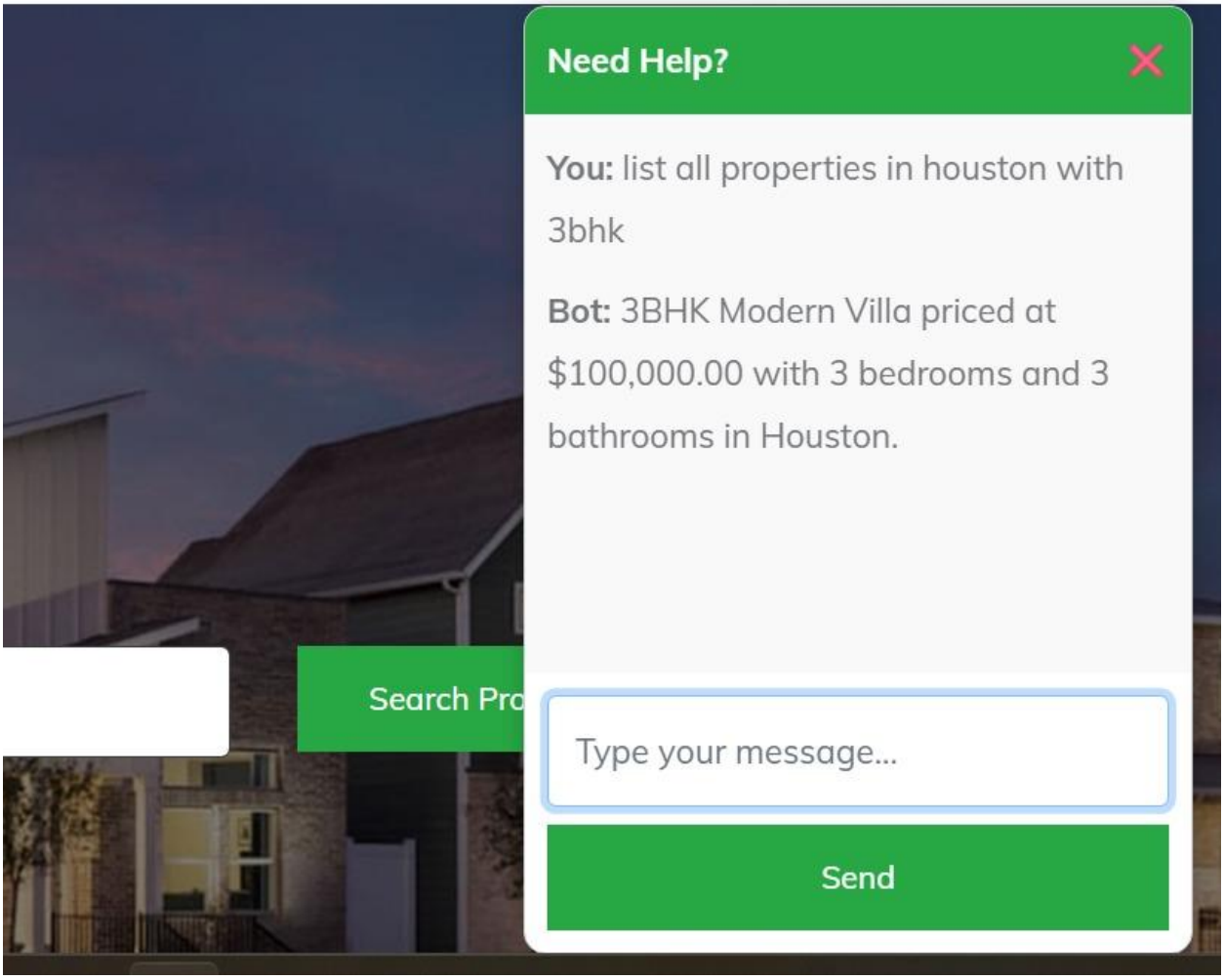

Fig 8: Chat-Bot

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8.Conclusion:

Through this project, we were able to design, plan and implement the Real Estate Property Management System (REPMS) catering to address the continuous needs of the real estate market. Our work focused on implementing and having a seamless transaction process by supporting multiple user roles and including sections for property listings, appointments, favorites, offers and contracts, secure payments and more. We emphasized data integrity, scalability, and usability by carefully structuring our relational database and defining clear business rules. We also incorporated a chatbot using an LLM to generate queries and communicate with the users via the front end of the system. This project helped us demonstrate and practice our understanding of database concepts and their practical application while also aiding us to learn and build ability in collaborating effectively as a team to solve real-world tasks.

9.Demonstration:

Watch our demonstration in the link provided on YouTube:

https://youtu.be/Yz888hRvTgs

10.Project Code :

https://github.com/bhandroid/RealEstate1

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11.References:

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[7] "Entity Relationship Diagram (ERD) Tutorial," Lucidchart, [Online]. Available: https://www.lucidchart.com/pages/er-diagrams. [Accessed: May 2, 2025].

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Team and Contributions:

S.No	Name	Contribution
1	Anudeep Bathina	ER diagram review, password encryption implementation, Agent Module, presentation consolidation, report documentation
2	Shashank Arjun Dandu	Team lead, Database design, normalization, testing queries, UI/UX design, Backend logic design, Presenter
3	Abhinandhan Reddy Gangula	Database design, business rules, SQL schema creation, integrity constraint design, Chat bot using LLM, Presenter
4	Pravallika Kilari	Backend logic design, Buyer module design, admin panel schema, presentation consolidation
5	Ananya Kura	ER diagram, testing queries, Admin module and relationship mapping, report documentation
6	Chandana Obulampalli	ER diagram creation, business rules, SQL schema creation, navigation flow, property listing UI, Seller module
7	Bhanu Prakash Ramineni	UI/UX design, Backend logic design, User authentication logic, Password security and payment gateway implementation, Buyer, Seller & Agent Modules, Presenter
8	Greeshma Geethika Rampam	Database design, normalization, testing queries, Presentation preparation and final report consolidation
9	Adi Karthikeya S B	ERD design in Draw.io, database connectivity testing, report documentation
10	Shanmukha Sai Vemulapalli	Database design, UI/UX design, appointment scheduling module, Backend logic design, Presenter